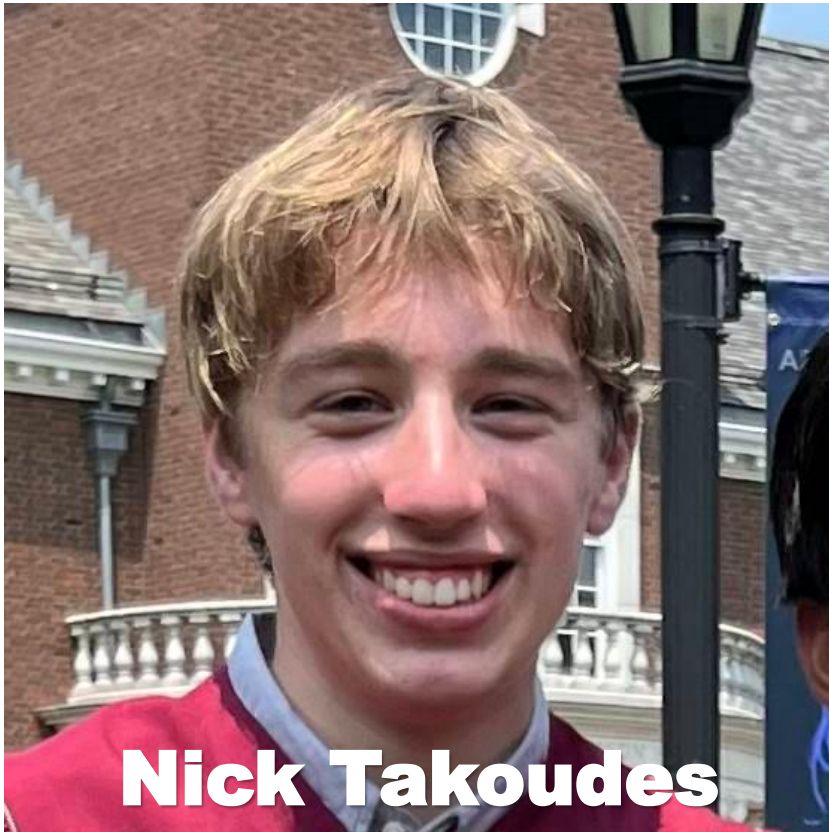


The Fourth Observable



Nick Takoudes

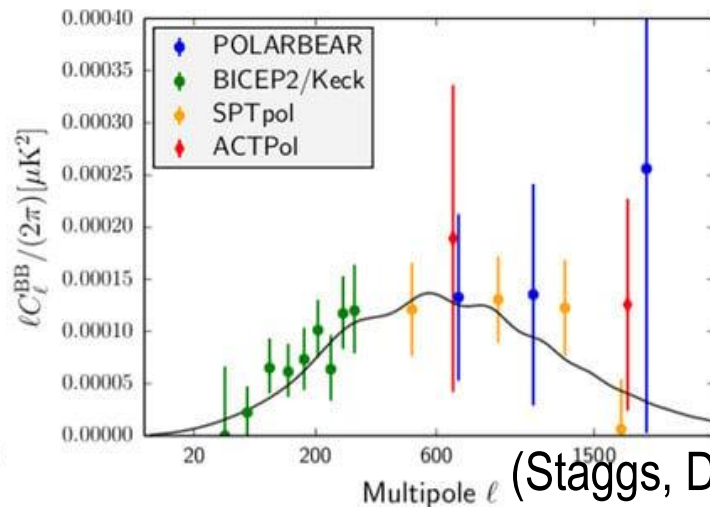
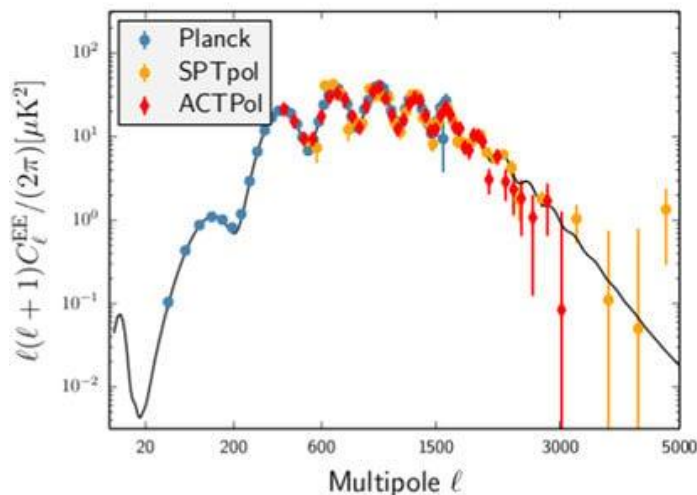
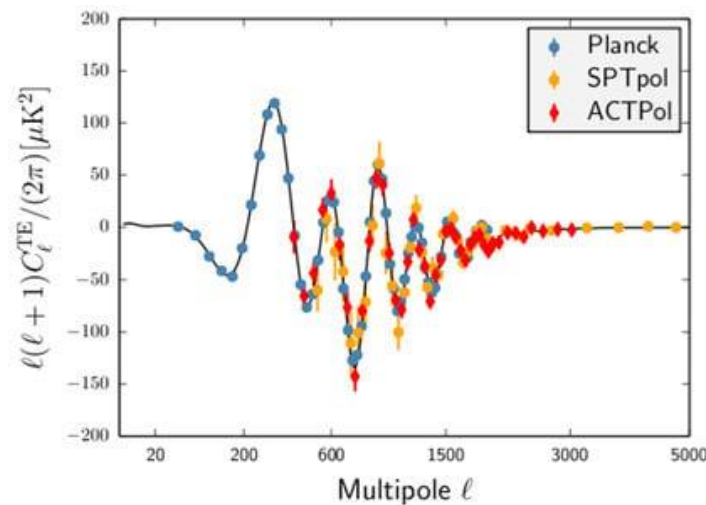
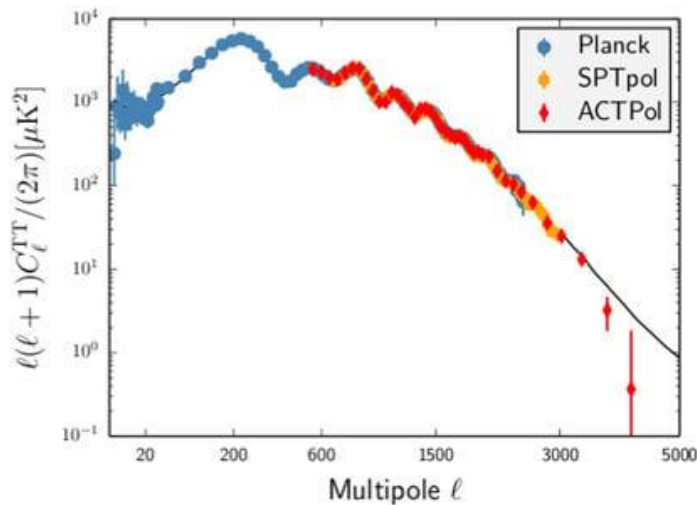


Hanjue Zhu

Nick Gnedin

The Three Observables (+X)

- Our beloved CMB, in its full glory.



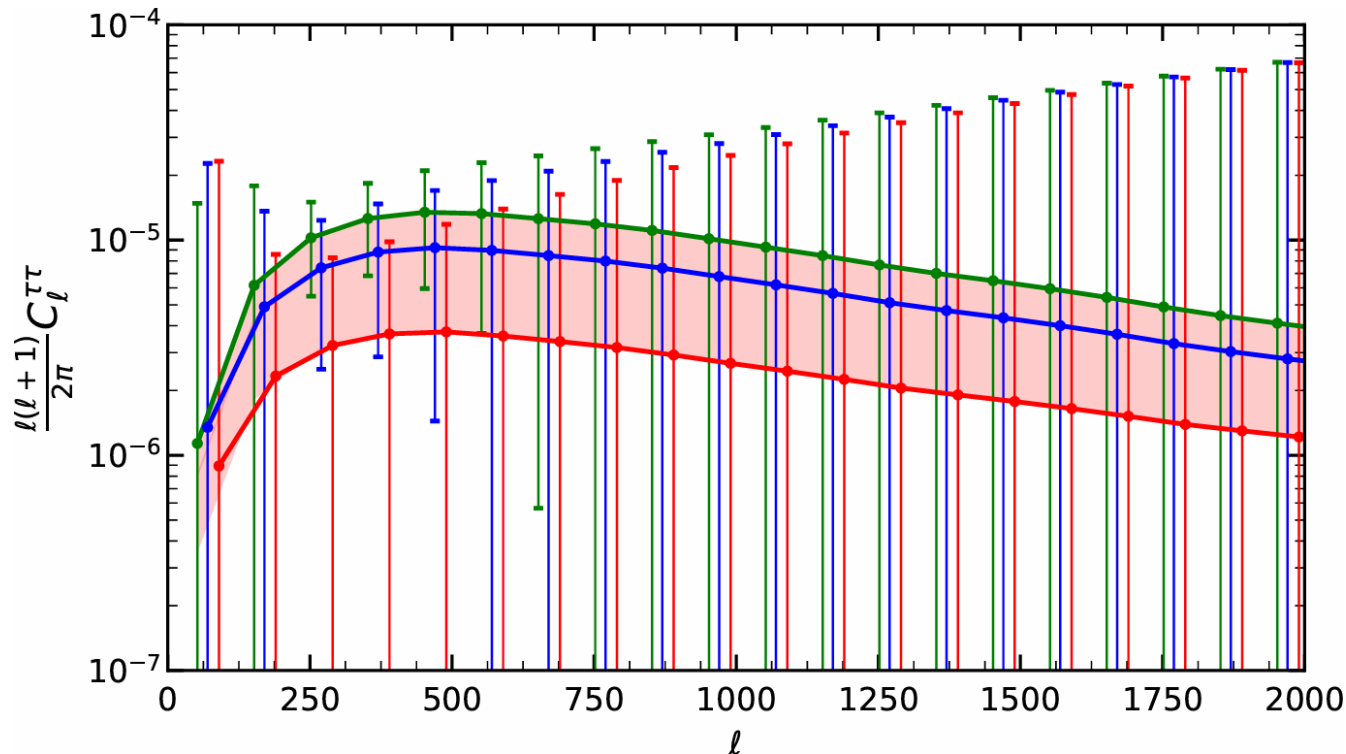
(Staggs, Dunkley, Page 2018)

The Three Observables (+X)

- The CMB comes from the surface of the last scattering.
- Late universe affects the CMB in 3 distinct ways:
 1. The Integrated Sachs-Wolfe effect modifies anisotropies on large scales.
 2. Gravitational lensing distorts small-scale fluctuations (the third observable).
 3. Cosmic reionization introduces the non-zero optical depths to Compton scattering and suppresses fluctuations on all scales.
- There is the fourth observable, though.

Reionization and the CMB

- Reionization introduces additional scatterings. This is quantified by the reionization optical depth τ_{CMB} .
- Sorry, Michael, it is not “just one number”.
- Dvorkin & Smith (2009): $\tau = \tau(\vec{n})$



(Roy++ 2022)

What Fluctuates?

- The reionization optical depth is the line-of-sight integral of the (free-) *electron density*:

$$\tau(\vec{n}) = \sigma_T \int n_e dr$$

- The electron density fluctuates because (a) *ionization is inhomogeneous*, and (b) the *density fluctuates*.
- Hence, there are two contributions: $n_e = x_e n_b$

$$\delta n_e = (x_e - \overline{x_e}) n_b + \overline{x_e} (n_b - \overline{n_b})$$

“Patchy reionization”

“Density fluctuations”

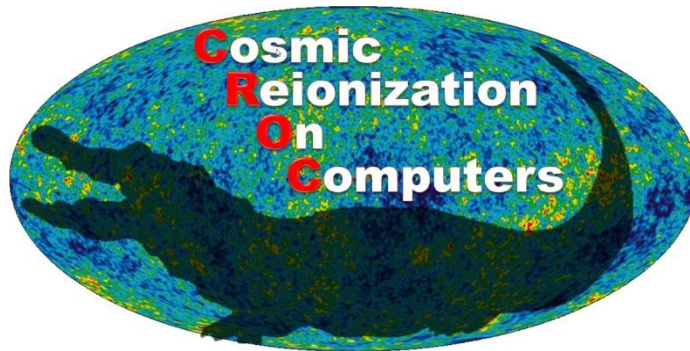
Modeling Reionization

- There are many useful ways to model reionization:
 - (Semi-)Analytical models (MHR 2000, FZH 2004, ...)
 - Semi-numerical schemes (21cmFAST, ARTIST, AMBER, ...)
 - DMO + SAM[ish] (ASTRAEUS/DELPHI, DRAGONS, GRIZZLY, ...)
 - Partially-coupled / RT-in-postprocessing simulations (Astrid, MassiveBlack-II, C2-Ray, Illustris, ...)
 - *Fully-coupled simulations*
- There is no good “golden solution”; all other methods are calibrated against fully-coupled simulations.

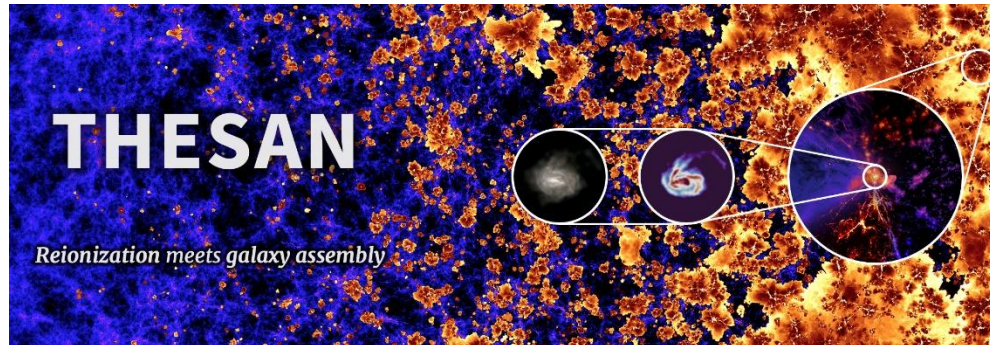
The Battle of Giants

(second generation simulations)

- CROC



- Thesan



- CoDa (Cosmic Dawn)
- Lumina (just announced)

Simulating Reionization: The Scorecard

	$M_{\text{box}}/M_{\text{cell}}$	L (cMpc)	Δx (pc)
CoDa	8192 ³ !	94	1,700 (unigrid)
CROC	2048 ³	117	100 (~400 Mh)
Thesan	2100 ³	96	300 ?
Lumina	6000 ³	500 !	?

Why Am I Giving This Talk?

THE ASTROPHYSICAL JOURNAL, 551:3–14, 2001 April 10

© 2001. The American Astronomical Society. All rights reserved. Printed in U.S.A.

Ⓔ

SECONDARY COSMIC MICROWAVE BACKGROUND ANISOTROPIES FROM COSMOLOGICAL REIONIZATION

NICKOLAY Y. GNEDIN

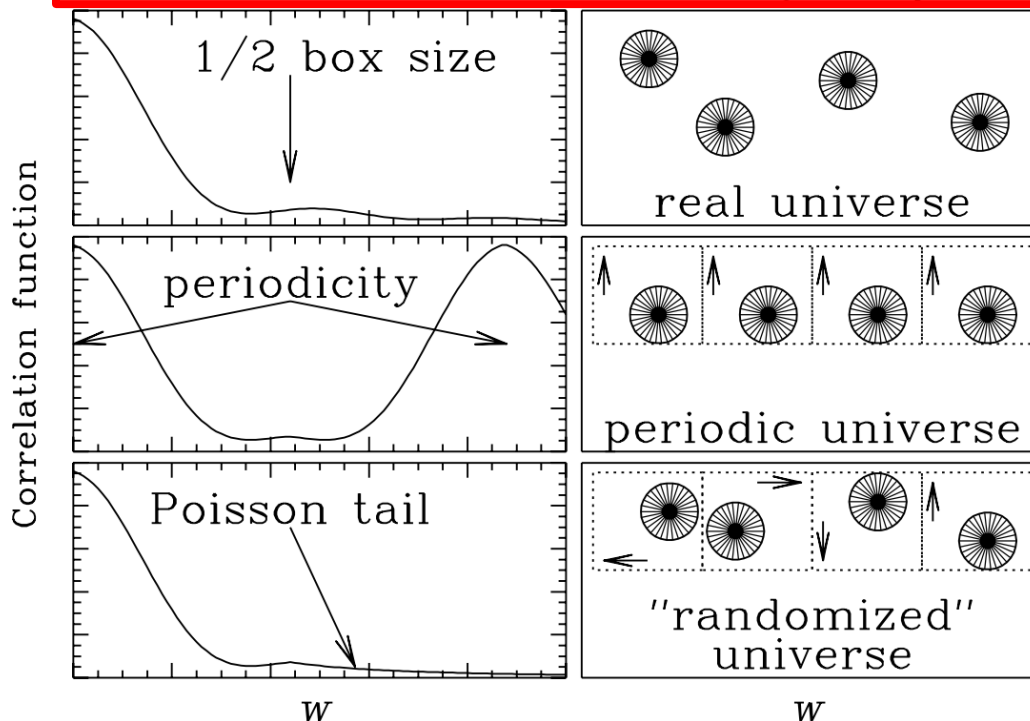
Center for Astrophysics and Space Astronomy, University of Colorado, Campus Box 389, Boulder, CO 80309; gnedin@casa.colorado.edu

AND

ANDREW H. JAFFE

Center for Particle Astrophysics, University of California, Berkeley, CA 94720; jaffe@cfpa.berkeley.edu

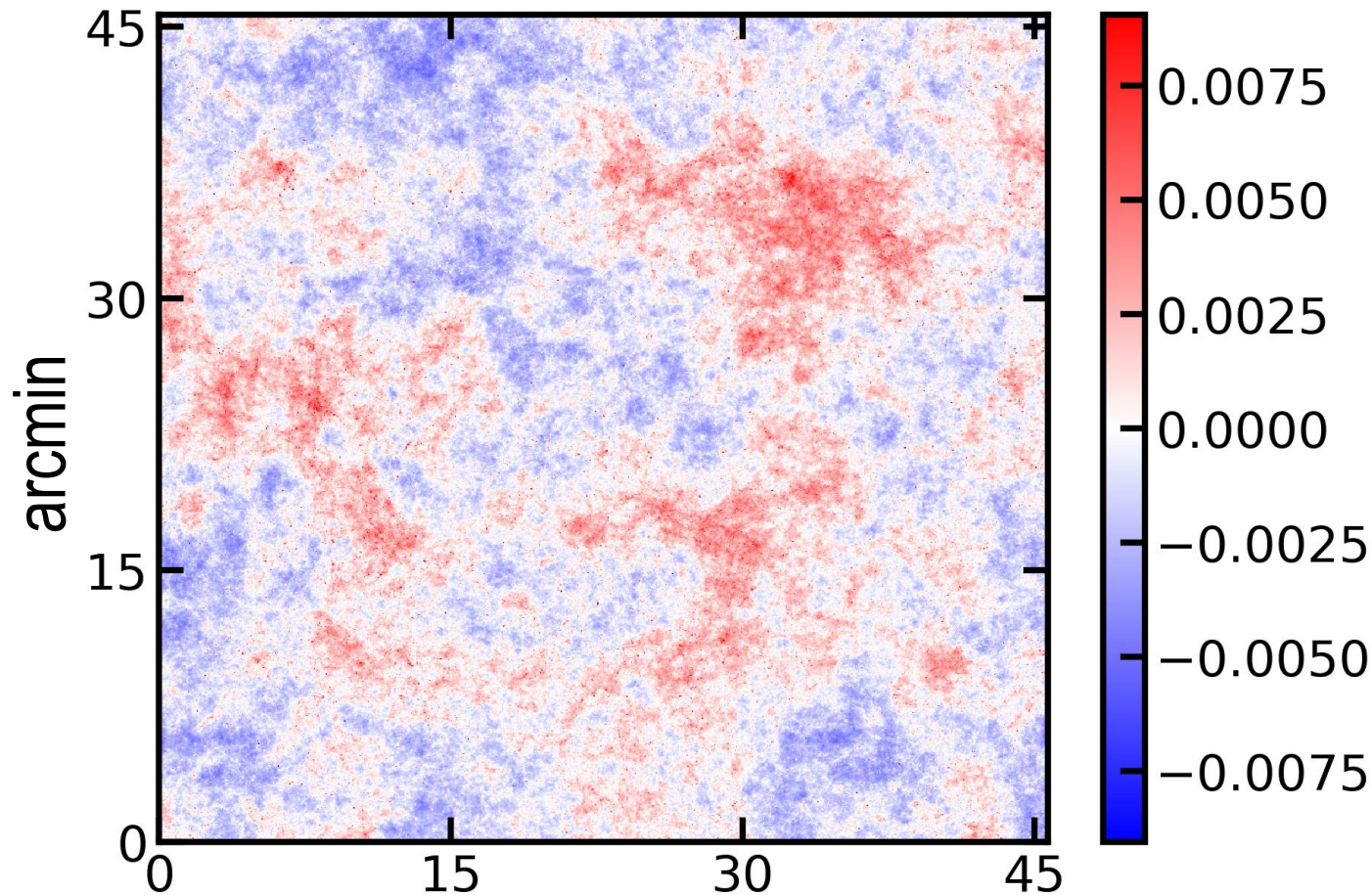
Received 2000 August 29; accepted 2000 November 16



- Building a light cone after the simulation is completed is a non-trivial task.
- I learned that trick from Andrew a long time ago...

τ_{CMB} Is Not “Just One Number”

- Fluctuations in τ_{CMB} are $\sim$ lognormal, with a high-density tail (CGM in halos).



What Fluctuates?

- The reionization optical depth is the line-of-sight integral of the (free-) *electron density*:

$$\tau(\vec{n}) = \sigma_T \int n_e dr$$

- The electron density fluctuates because (a) *ionization is inhomogeneous*, and (b) the *density fluctuates*.
- Hence, there are two contributions: $n_e = x_e n_b$

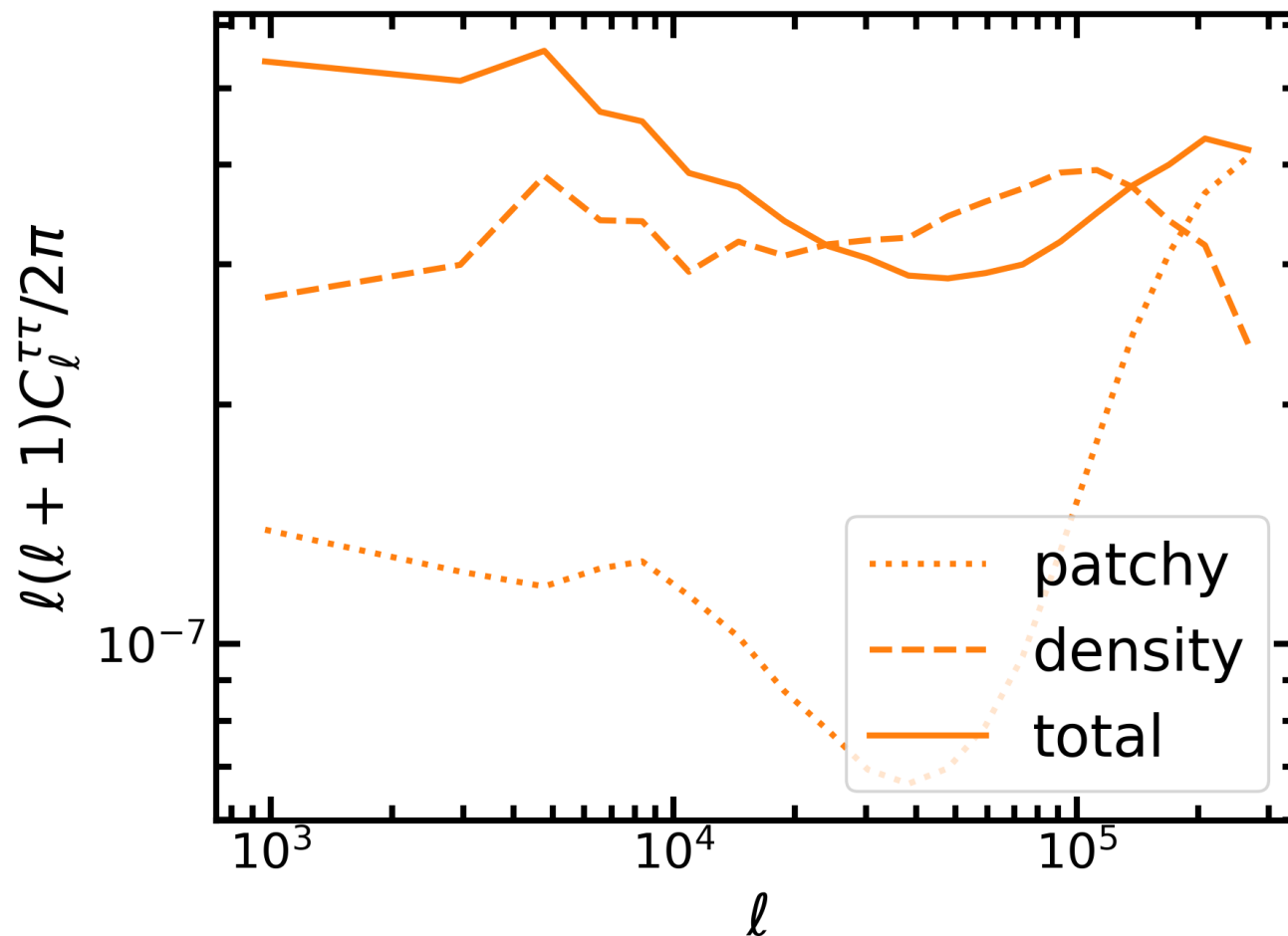
$$\delta n_e = (x_e - \overline{x_e}) n_b + \overline{x_e} (n_b - \overline{n_b})$$

“Patchy reionization”

“Density fluctuations”

Starting Simple

- The power spectrum is the most familiar statistic: starting with it.



Starting Simple

- There is one problem with reionization simulations: none of them go to $z=0$.
- After reionization $x_e = 1$.

$$\delta n_e = \cancel{(x_e - \bar{x}_e)} n_b + \bar{x}_e (n_b - \bar{n}_b)$$

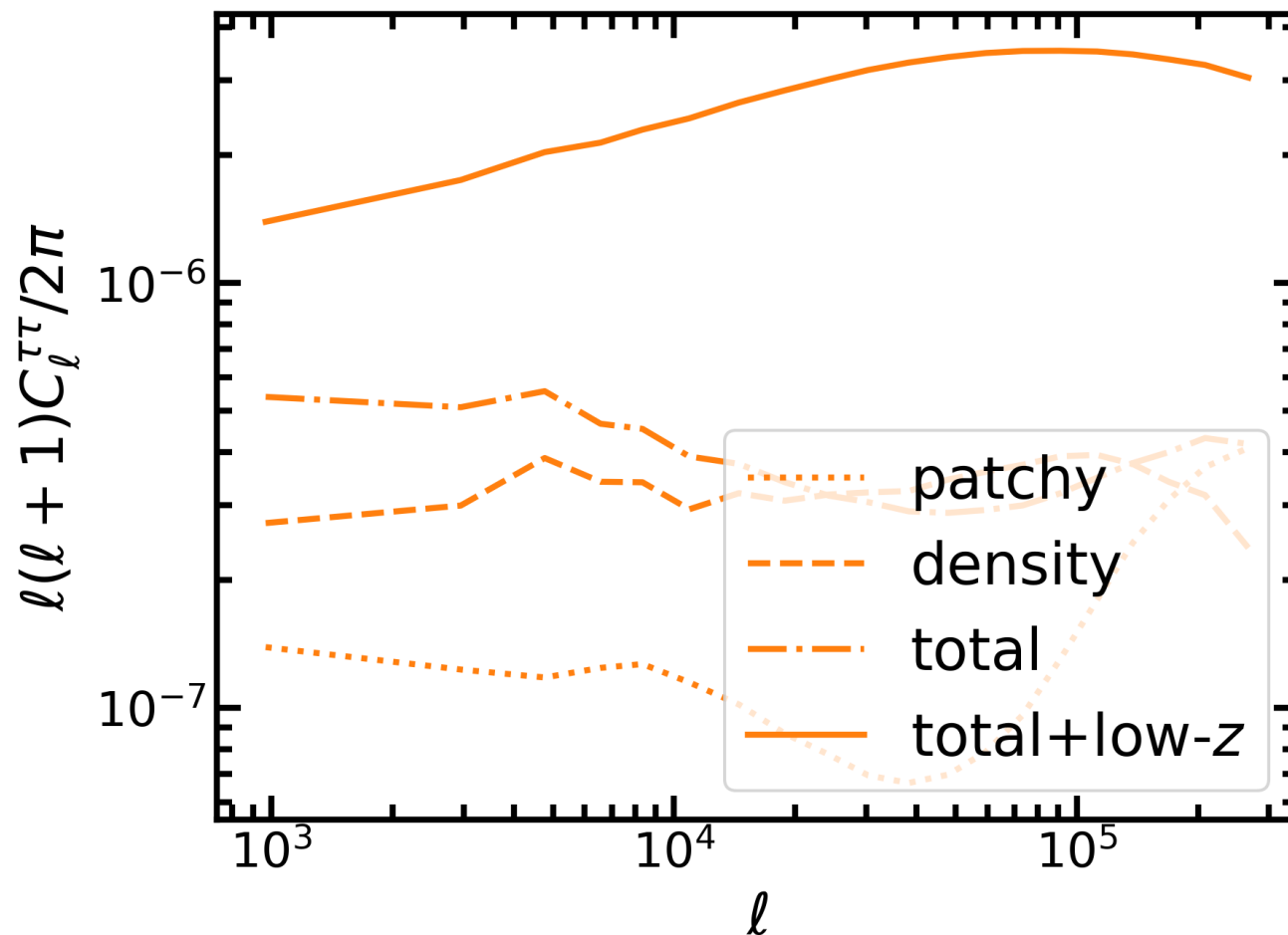
Still there 

- Limber's approximation:

$$C_\ell^{\tau\tau} = \sigma_T^2 \int \frac{dx}{ax^2} P_{\delta n_e \delta n_e} \left(x, k = \frac{\ell}{x} \right)$$

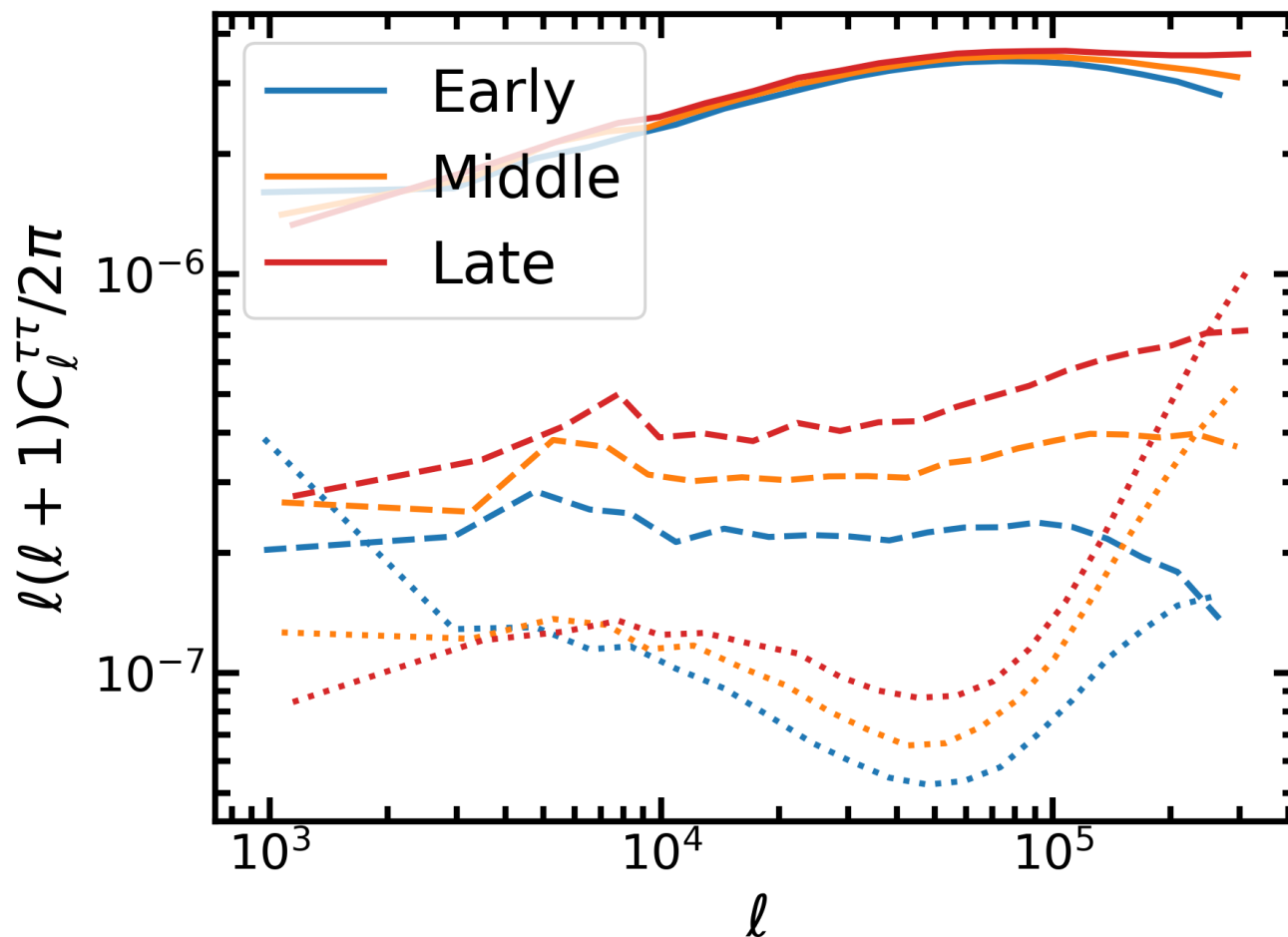
Starting Simple

- The bummer: the low- z contribution dwarves all contributions from reionization. ☹



Starting Simple

- There is dependence on the reionization history, but it is washed out by the low- z contribution. ☹



Conclusions

- Fluctuations in the reionization optical depths are the fourth CMB observable.
- They have been expected to serve as an independent probe for studying reionization.
- Unfortunately, density fluctuations after reionization also contribute to fluctuations in the reionization optical depth, and they dominate fluctuations from reionization in the power spectrum (structure keeps growing, the path length is larger).
- Perhaps, higher-order statistics will be more sensitive...